

Parking Users

Students – Access to their own accounts and must self-manage their fines and account. Can register multiple personal vehicles and get a permit which lasts per semester.

Staff – Have everything as students. Also have priority premium lots and priority parking levels.

Visitors – Guests can register for specific time frame parking times in designated parking locations.

Admin – Oversees the entire database and determines the overall picture of the database and who has access to certain actions.

Role-Permission Matrix

Feature / Action	Student	Faculty	Visitor	Admin
Register Multiple Vehicles	Yes	Yes	No	Yes
Purchase Long-term Permit	Yes	Yes	No	Yes
Reserve Visitor Spot	No	No	Yes	Yes
Pay Personal Fines	Yes	Yes	Yes	Yes
Issue/Modify Tickets	No	No	No	Yes

Modules – Each module will include a group of related tables in the ER diagram

- **Account and Profile Module:** Manages users, vehicles, and roles and makes sure vehicles have unique license plate numbers.
- **Access and Zone Module:** Manages lots, zones, and spots and make sure students can't park in staff section and alike.
- **Permit and Reservation Module:** Handles permits (for users such as students and staff) and reservations (for visitors)
- **Enforcement Module:** It processes data from the database and compares it against the permits and reservations.

Business Rules

1. Users can register multiples vehicles but with different license plate numbers.

- a. Enforced by (A). This is a UNIQUE constraint on the license plate column and a foreign key linking to the user_id (for identification of the exact user)
2. Each parking space must be identified with the specific building and parking space number.
 - a. Enforced by (A). COMPOSITE PRIMARY KEYS constraint (building_id + space_number) for uniqueness for each spot.
3. Non-users are considered visitors and can register for parking spots at a specific building and parking space.
 - a. Enforced by (A). FOREIGN KEY in the Reservations table linking a visitor_id to space_id
4. The visitor register start time must be before the register end time.
 - a. Enforced by (A). CHECK constraint. It checks to make sure the end time is after the start time.
5. Fines are given to users and visitors if they don't park in the appropriate spot. Fines are also given if visitors stay longer than their designated end time.
 - a. Enforced by (B). Using a TRIGGER to calculate the time violation and insert a fine.
6. When cars are registered, the database updates the spot and matches it with the student's profile.
 - a. (B) Matching and updating are actions. A TRIGGER runs to change the spot status when a registration is detected.
7. Parking spots cannot be booked for overlapping times for some or different users.
 - a. (B) Using TRIGGER to block overlapping changes.
8. When a car enters a parking spot, the database registers it and updates the user table side to the appropriate car for the appropriate user.
 - a. (B) When spot is marked as occupied, a TRIGGER is used to find the car and update the current location in the user and car records.
9. If a visitor tries to park in a section they are not allowed to park in, the system will insert a ticket into their account if they have enrolled before, if not it sends a notification to UMBC police which then finds different ways to send the fines to the driver.
 - a. (B) TRIGGER. Using If-else statements to check if user exists and sending notifications/emails to UMBC police.
10. The system doesn't allow visitors to register a car.
 - a. (B) TRIGGER used to give custom error messages.

Concurrency

If two users try to reserve the same spot at the same time, the system stops both from reserving and leaves a message saying something like, “Too many users trying to enroll at the same time, try again later, in a little bit.”

Design Rationale

I have different modules for the database because I need to separate the different tables by what they are related to. It's important to separate the user account tables from the permits, lots, and reservations, etc. We need to be able to separate staff from students as they have different abilities and priorities and should therefore be separated. For the rules table above, I made sure I gave admin the ability every member (student, staff, ...) has so they can modify things accordingly. I will allow visitors to reserve as much time as they need with the restraint of 24 hours and it will be free.

ER Diagram from Mermade.js

